

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**

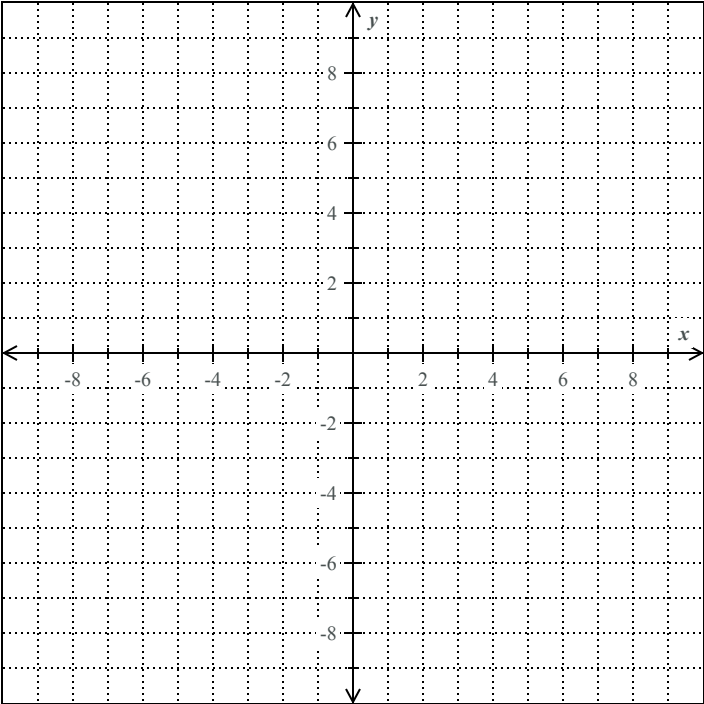
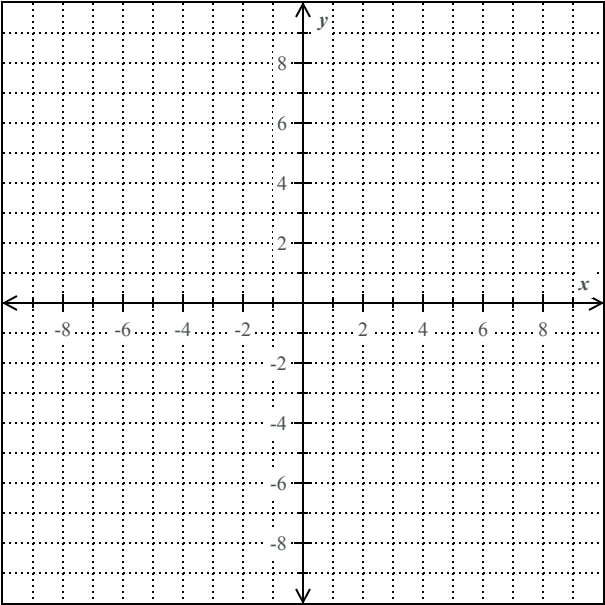
Number of Questions: **10**

Instructor Name: **Mamani Velasco, Ruben**

Question 1 of 10

Graph the ellipse.

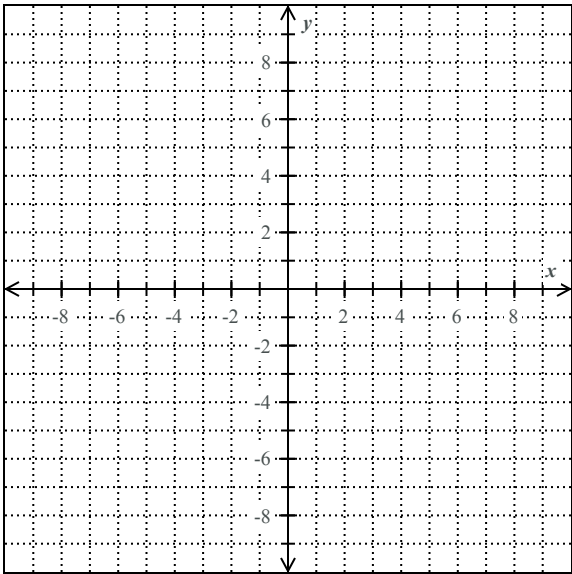
$$\frac{(x + 2)^2}{9} + (y - 5)^2 = 1$$



Question 2 of 10

Graph the ellipse.

$$4x^2 + 9y^2 = 36$$



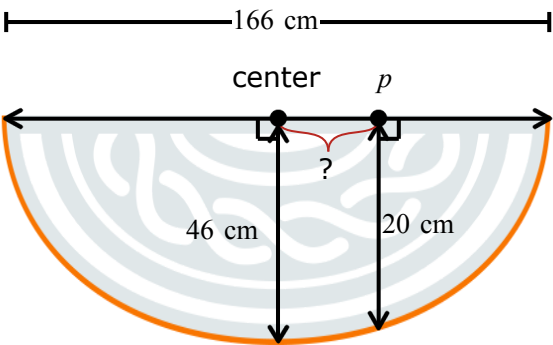
Question 3 of 10

Find an equation of the ellipse that has center $(-5, 3)$, a minor axis of length 2, and a vertex at $(-5, -3)$.

Question 4 of 10

The curved edge of the door mat below is half an ellipse, a "semi-ellipse". As shown in the figure below, the flat edge of the door mat measures 166 cm and the distance from the center (on the flat edge) to the curved edge is 46 cm. The distance from point p to the curved edge is 20 cm. Find the distance from p to the center.

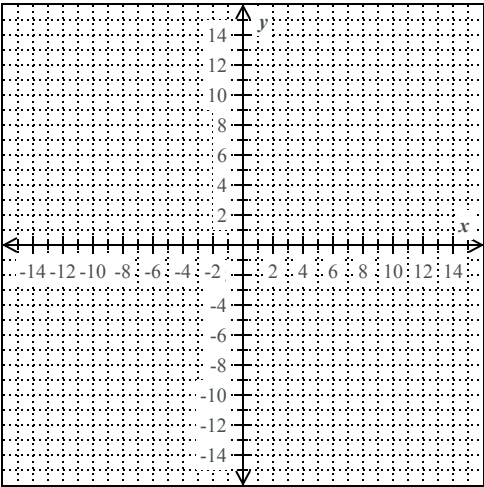
Round your answer to the nearest hundredth. Do not round any intermediate computations.



Question 5 of 10

Graph the hyperbola.

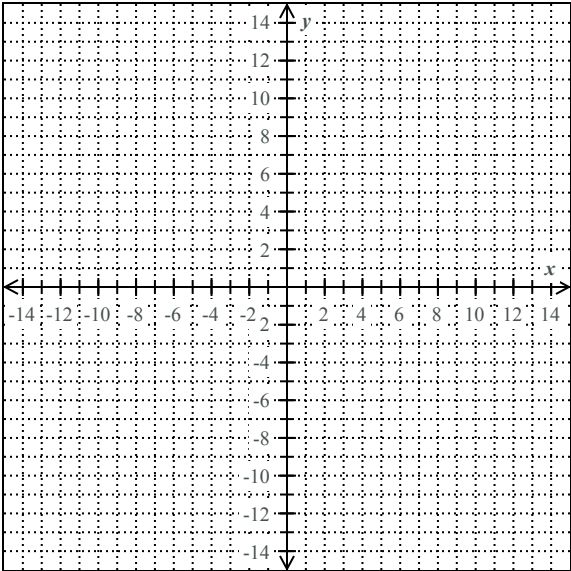
$$\frac{(x - 6)^2}{25} - \frac{(y - 2)^2}{16} = 1$$



Question 6 of 10

Graph the hyperbola.

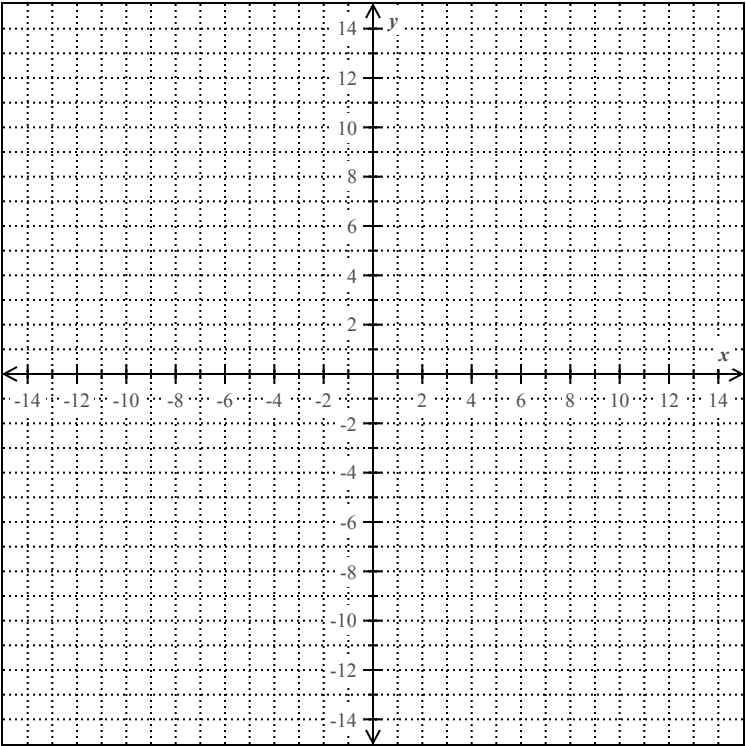
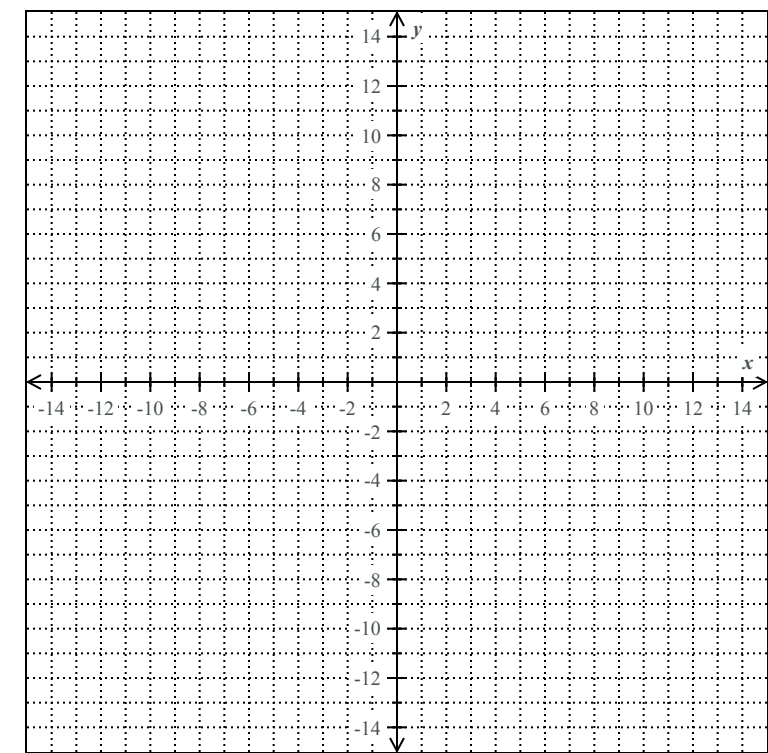
$$x^2 - 25y^2 = 25$$



Question 7 of 10

Graph the hyperbola.

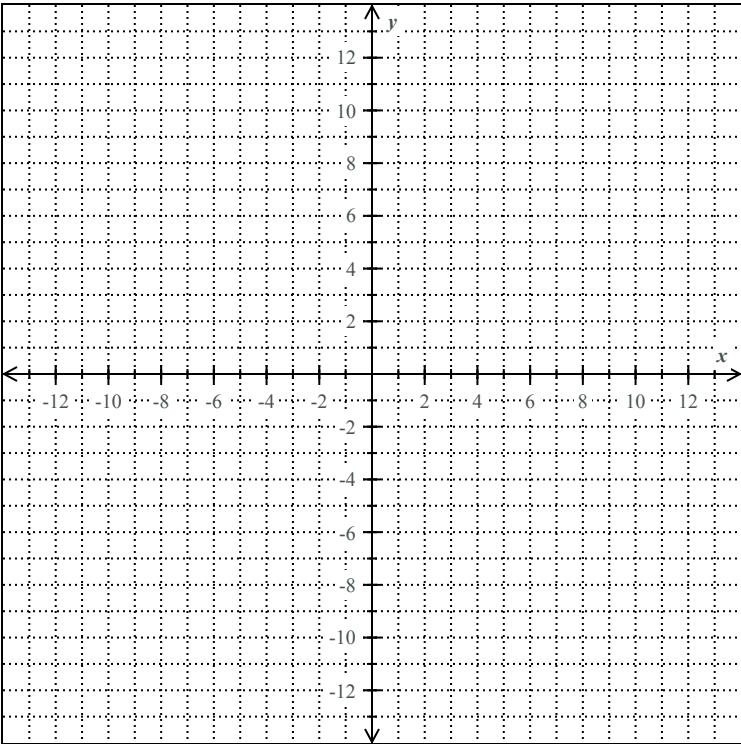
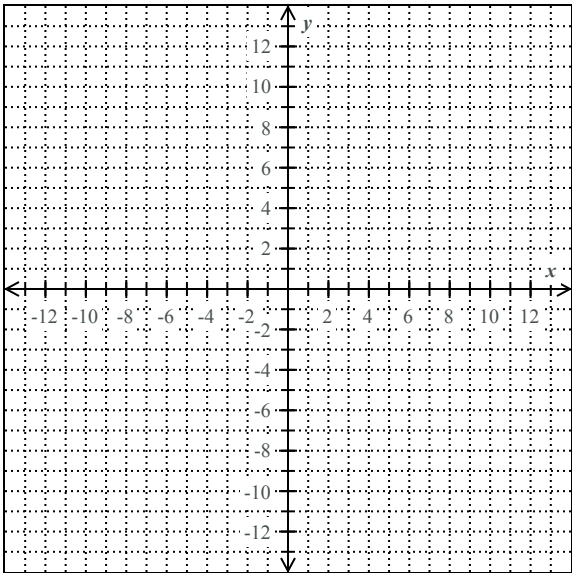
$$9x^2 - 4y^2 - 54x - 24y + 9 = 0$$



Question 8 of 10

Graph the parabola.

$$y^2 - 8y - x + 13 = 0$$

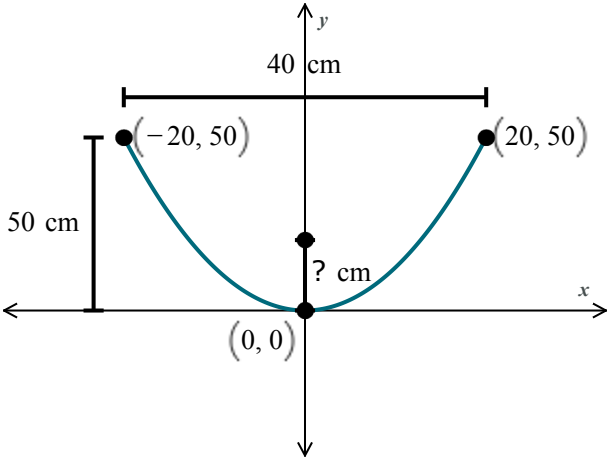
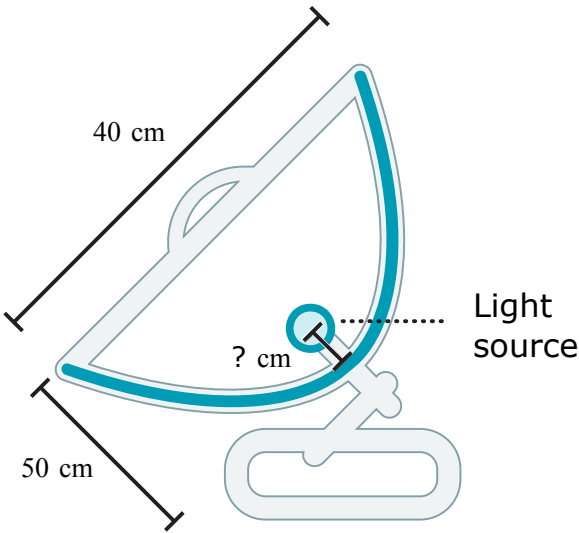


Question 9 of 10

Find an equation of the parabola with vertex $(-1, 2)$ and focus $(-1, -5)$.

Question 10 of 10

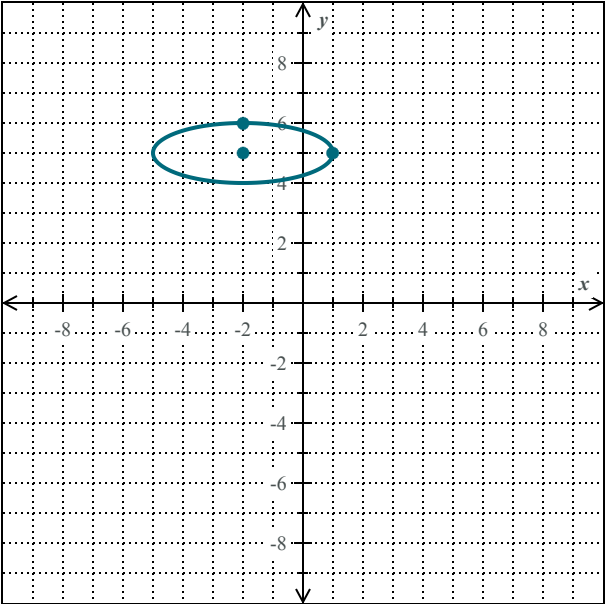
A lamp has the shape of a parabola when viewed from the side. The lamp is 40 centimeters wide and 50 centimeters deep. How far is the light source from the bottom of the lamp if the light source is placed at the focus?



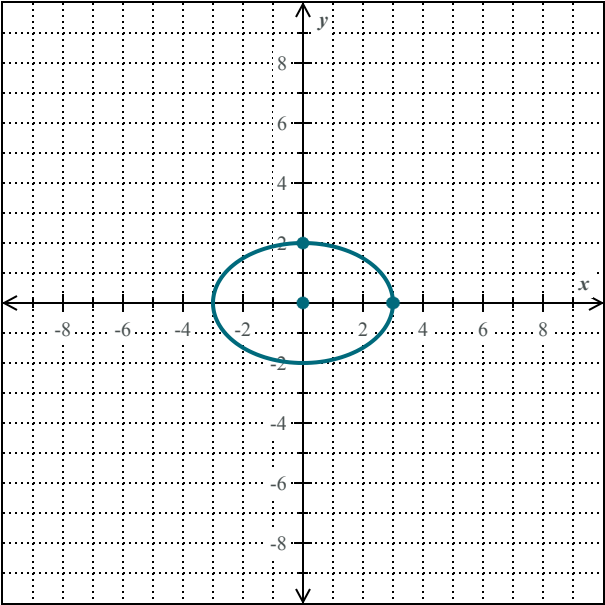
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Number of Questions: **10**

Question 1 of 10



Question 2 of 10



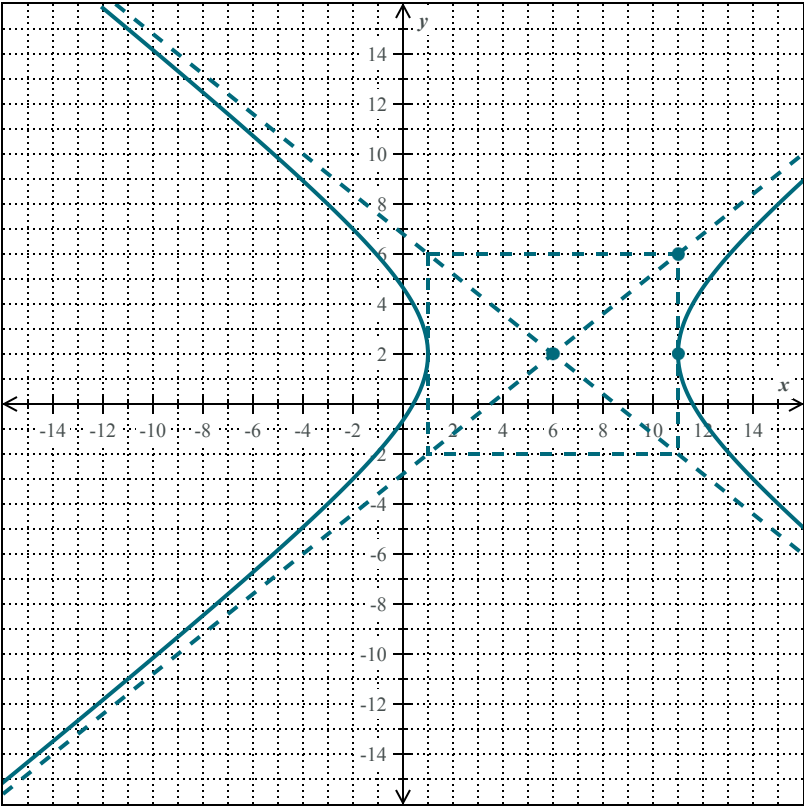
Question 3 of 10

$$\frac{(x+5)^2}{1} + \frac{(y-3)^2}{36} = 1$$

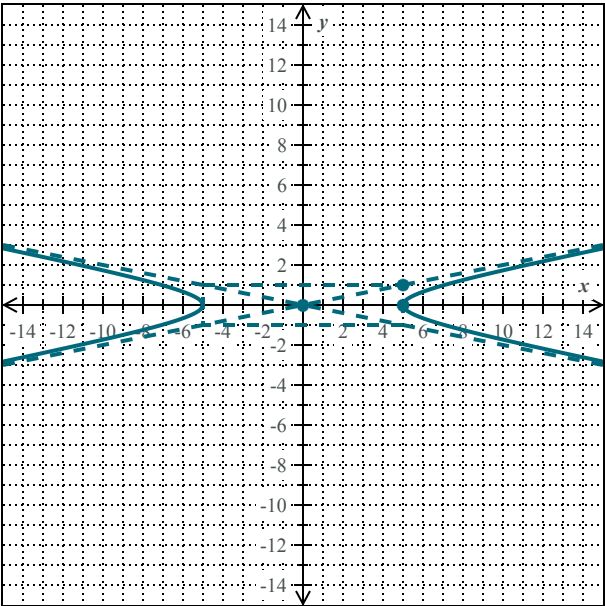
Question 4 of 10

74.74 cm

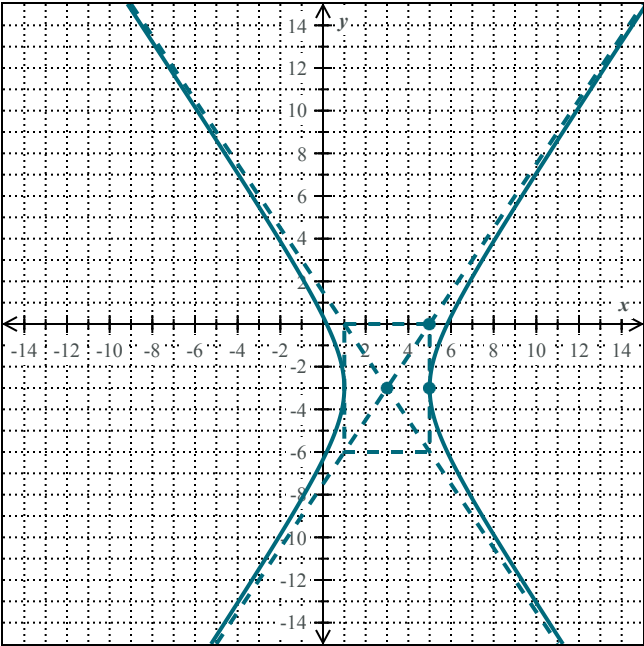
Question 5 of 10



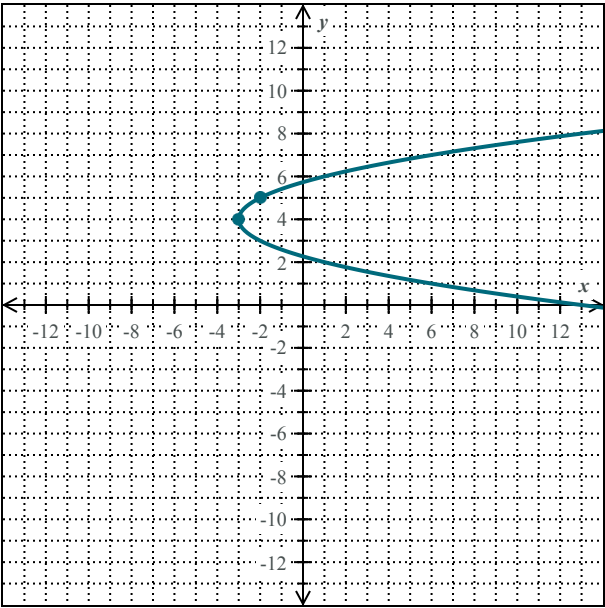
Question 6 of 10



Question 7 of 10



Question 8 of 10



Question 9 of 10

$(x + 1)^2 = -28(y - 2)$

Question 10 of 10

2 cm